CORE VALUES | QUALITY PRODUCTS | SUPERIOR SERVICE

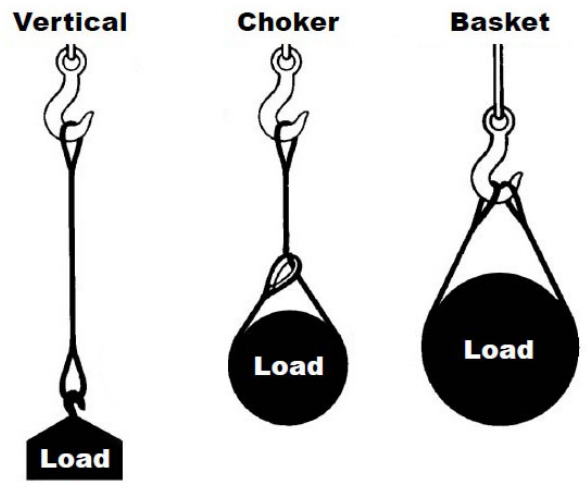
CORE® LIFTING PRODUCTS

CORE-TEX®

POLYESTER ROUNDSLINGS

CoreLifting.com

General Information
Basic Hitches



VERTICAL, or straight, attachment is simply using a sling to connect a lifting hook to a load. Full rated lifting capacity of the sling may be utilized but must not be exceeded. A tagline should be used to prevent load rotation, which may damage a sling. When two or more slings are attached to the same lifting hook, the total hitch becomes, in effect, a lifting bridle, and the load is distributed equally among the individual slings.

CHOKER hitches reduce lifting capability of a sling since this method of rigging affects ability of the wire rope components to adjust during the lift. A choker is used when the load will not be seriously damaged by the sling body — or the sling damaged by the load, and when the lift requires the sling to snug up against the load.

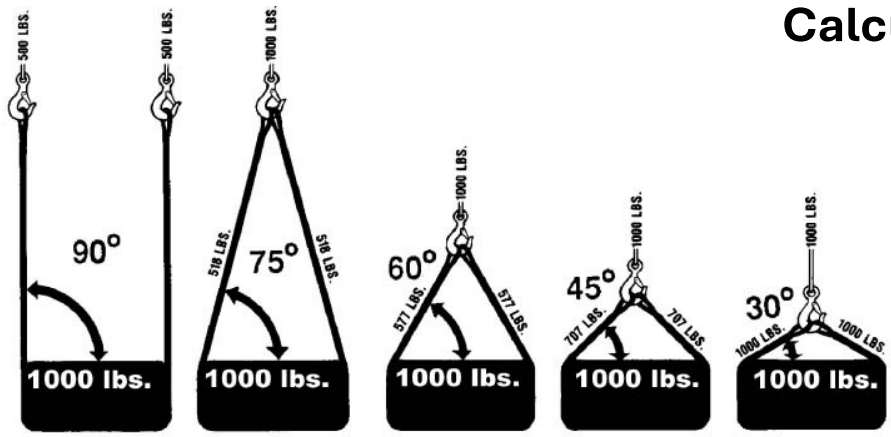
The diameter of the bend where the sling contacts the load should keep the point of choke against the sling BODY — never against a splice or the base of the eye. When a choke is used at an angle of less than 120 degrees (see next page), the sling-rated capacity must be adjusted downward.

A choker hitch should be pulled tight before a lift is made — NOT PULLED DOWN DURING THE LIFT. It is also dangerous to use only one choker hitch to lift a load which might shift or slide out of the choke.

BASKET hitches distribute a load equally between the two legs of a sling ... within limitations described above.

General Information
Load Calculating

Calculating the Load on Each Leg of a Sling



As the included angle between the legs of a sling decreases, the load on each leg increases. The effect is the same whether a single sling is used as a basket or two slings are used with each in a straight pull, as with a 2-legged bridle.

Anytime pull is exerted at an angle on a leg—or legs—of a sling, the load per leg can be determined by using the data in the table above. Proceed as follows to calculate this load—and determine the rated capacity required of the sling, or slings, needed for a lift.

1.First, divide the total load to be lifted by the number of legs to be used. This provides the load per leg if the lift were being made with all legs being vertically.

2.Determine the angle between the legs of the sling and the horizontal.

3.Then MULTIPLY the load per leg (as computed above) by the Load Factor for the leg angle being used (from the table at the bottom) – to compute the ACTUAL LOAD on each leg for this lift and angle. THE ACTUAL LOAD MUST NOT EXCEED THE RATED SLING CAPACITY.

Thus, in the above drawing (sling angle at 60°): 1000 ÷ 2 = 500 (Load Per Leg if a vertical lift) 500 x 1.154 = 577 lbs. – ACTUAL LOAD on each leg at the 60° included angle being used.

In the above drawing (sling angle of 45°): 1000 ÷ 2 = 500 (Load Per Leg if a vertical lift) 500 x 1.414 = 707 lbs. = ACTUAL LOAD on each leg at the 45° horizontal angle being used.

General Information
Bridle Legs – Angle Effects

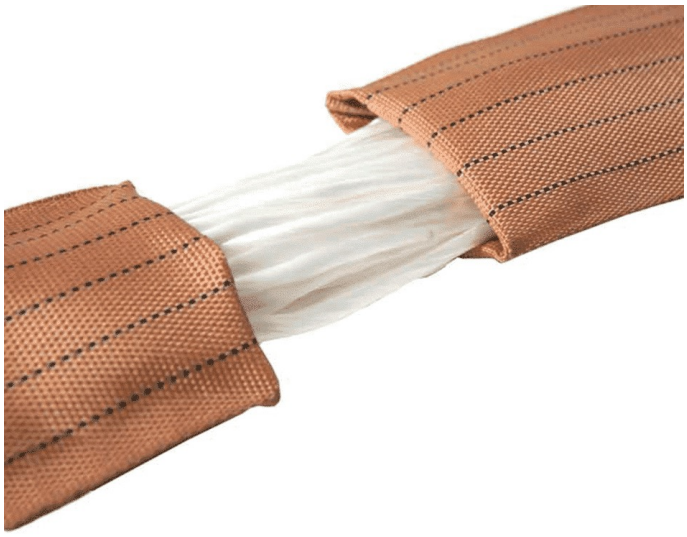
Angles of Bridles

The horizontal angle of bridles with 3 or more legs is measured the same as the horizontal sling angle of 2-legged hitches. In this case, where a bridle designed with different leg lengths results in horizontal angles, the leg with the smallest horizontal angle will carry the greatest load. Therefore, the smallest horizontal angle is used in calculating actual leg load and evaluating the rated capacity of the sling proposed.

Leg Angle (Degrees)	Load Factor
90	1.000
85	1.003
80	1.015
75	1.035
70	1.064
65	1.103
60	1.154
55	1.220
50	1.305
45	1.414
40	1.555
35	1.743
30	2.000

General Information
About Polyester Roundslings

Polyester roundslings are a polyester fiber lifting sling that can be used in a vertical, choker, or basket hitch configuration. The sling consists of a group of twisted polyester fiber strands that are encapsulated in a polyester woven tube that is colored to indicate capacity.



These slings are strong, extremely flexible and lightweight. These attributes make them very suitable in the construction industry, in use under overhead cranes and countless other applications.

Color Capacities

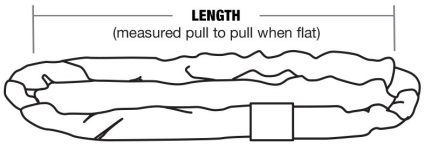
Purple 2,600 lbs Vertical
Green 5,300 lbs Vertical
Yellow 8,400 lbs Vertical
Tan 10,600 lbs Vertical
Red 13,200 lbs Vertical
White 16,800 lbs Vertical
Blue 21,200 lbs Vertical
Orange 25,000 lbs Vertical
Grey 31,000 lbs Vertical
Orange 40,000 lbs Vertical
Brown 53,000 lbs Vertical
Olive 66,000 lbs Vertical
Black 90,000 lbs Vertical

Polyester Roundslings
Sling Capacities

CORE-TEX®

Single Sling

Color Code	Rated Capacity - Lbs Vertical	Choker	Basket
Purple	2,600	2,100	5,200
Green	5,300	4,240	10,600
Yellow	8,400	6,700	16,800
Tan	10,600	8,500	21,200
Red	13,200	10,600	26,400
White	16,800	13,400	33,600
Blue	21,200	17,000	42,400
Orange	25,000	20,000	50,000
Grey	31,000	24,800	62,000
Orange	40,000	32,000	80,000
Brown	53,000	42,400	106,000
Olive	66,000	52,800	132,000
Black	90,000	72,000	180,000



2-Leg Bridle

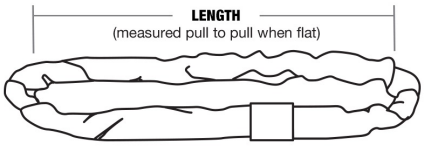
	Rated Capacity - Lbs			
	One Leg	All Legs		
Color Code	Vertical	60°	45°	30°
Purple	2,600	4,500	3,600	2,600
Green	5,300	9,100	7,400	5,300
Yellow	8,400	14,500	11,800	8,400
Tan	10,600	18,300	14,900	10,600
Red	13,200	22,800	18,600	13,200
White	16,800	29,100	23,700	16,800
Blue	21,200	36,700	29,900	21,200
Orange	25,000	43,200	35,200	25,000
Grey	31,000	53,700	43,800	31,000
Orange	40,000	69,200	56,400	40,000
Brown	53,000	91,800	74,900	53,000
Olive	66,000	114,300	93,300	66,000
Black	90,000	155,800	127,200	90,000

Polyester Roundslings
Sling Capacities

CORE-TEX®

3-Leg Bridle

	Rated Capacity - Lbs			
	One Leg	All Legs		
Color Code	Vertical	60°	45°	30°
Purple	2,600	6,700	5,500	3,900
Green	5,300	13,700	11,200	7,900
Yellow	8,400	21,800	17,800	12,600
Tan	10,600	27,500	22,400	15,900
Red	13,200	34,200	27,900	19,800
White	16,800	43,600	35,600	25,200
Blue	21,200	55,000	44,900	31,800
Orange	25,000	64,700	52,700	37,500
Grey	31,000	80,500	65,700	46,500
Orange	40,000	103,600	84,400	60,000
Brown	53,000	137,600	112,400	75,900
Olive	66,000	171,400	139,900	99,000
Black	90,000	233,800	190,800	135,000



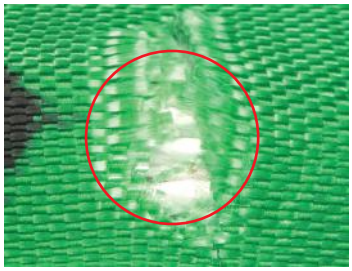
4-Leg Bridle

	Rated Capacity - Lbs			
	One Leg	All Legs		
Color Code	Vertical	60°	45°	30°
Purple	2,600	9,000	7,300	5,200
Green	5,300	18,300	14,900	10,600
Yellow	8,400	29,100	23,700	16,800
Tan	10,600	36,700	29,900	21,200
Red	13,200	45,700	37,300	26,400
White	16,800	58,200	47,500	33,600
Blue	21,200	73,400	59,900	42,400
Orange	25,000	86,500	70,500	50,000
Grey	31,000	107,300	87,600	62,000
Orange	40,000	138,400	112,800	80,000
Brown	53,000	183,600	149,900	106,000
Olive	66,000	228,600	186,600	132,000
Black	90,000	311,700	254,500	180,000

ROUNDSLING INSPECTION CRITERIA

The following photos illustrate some of the damage that occurs and indicates the sling must be taken out of service. For inspection frequency requirements, see the General Information section in this catalog.

Cuts to the Cover
(Internal Core Yarns Exposed)



WHAT TO LOOK FOR
Broken fi bers of equal length indicate that the sling has been cut. When core yarns are exposed, the damage to the yarns cannot be determined. The sling must be taken out of service or inspected by a Core Lifting expert.

TO PREVENT
Always protect synthetic slings from being cut by using cut protection.

SUGGESTED *LIFT-ALL* PRODUCT: *Edge Defender*™
See the Sling Protection section in this catalog for more sling protection options.

Cuts to the Cover
(Internal Core Yarns NOT Exposed)

WHAT TO LOOK FOR
The double-walled jacket protects the inner core yarns from damage. If the damage appears only to the outer jacket and does not expose the inner core yarns, the sling may remain in service (except chemical or heat damage). The sling may also be returned to Core Lifting for inspection and repair to the jacket.

TO PREVENT
Use the appropriate sling protection between the sling and all edges that come in contact with the sling.

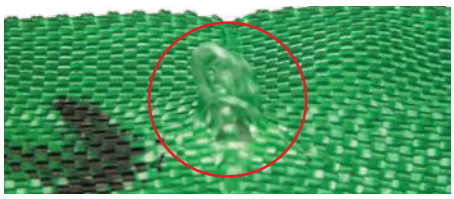
SUGGESTED *LIFT-ALL* PRODUCT: Wear Pad
See the Sling Protection section in this catalog for more sling protection options.

Holes, Snags, or Pulls

WHAT TO LOOK FOR
Punctures or areas where fibers stand out from the rest of the sling surface. Inspect the sling and remove from service if core yarn is exposed.

TO PREVENT
Avoid sling contact with protrusions, both during lifts and while transporting or storing.

SUGGESTED CORE LIFTING PRODUCT: Wear Pad
Contact Core Lifting for more sling protection options.



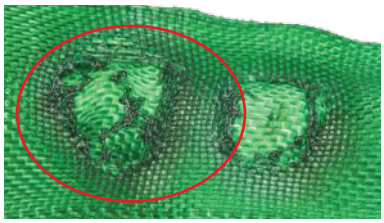
Knots

WHAT TO LOOK FOR
Knots compromise the strength of slings by not allowing all fibers to contribute to the lift as designed. Knots are rather obvious problems as shown here.

TO PREVENT
Never tie knots in slings.

ROUNDSLING INSPECTION CRITERIA

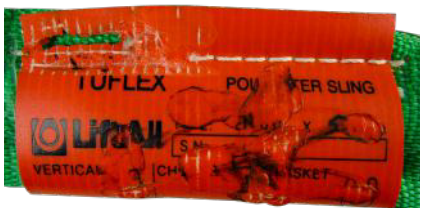
Heat / Chemical



WHAT TO LOOK FOR
Melted or charred fibers anywhere along the sling. Heat and chemical damage look similar and can damage sling fibers, compromising the sling's strength. Look for discoloration and/or fibers that have been fused together and may feel hard or crunchy. Slings showing heat or chemical damage must be removed from service.

TO PREVENT
Never use Core-Tex roundslings where they can be exposed to temperatures in excess of 200°F, or around chemicals without confirming that the sling material is compatible with the chemicals being used. For elevated temperatures up to 350°F.

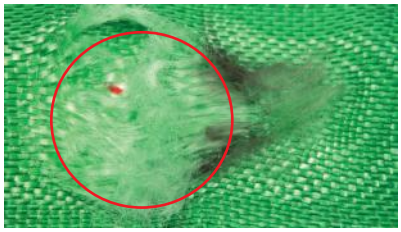
Illegible or Missing Tags



WHAT TO LOOK FOR
The information provided on the sling tag is important for knowing what sling to use and how it will function. If you cannot find or read all of the information on a sling tag, the sling must be taken out of service.

TO PREVENT
Never set loads down on top of slings or pull slings from beneath loads if there is any resistance. Load edges should never contact sling tags during the lift. Avoid paint or chemical contact with tags.

Abrasion



WHAT TO LOOK FOR
Areas of the sling that look and feel fuzzy indicate that the fibers have been broken by contact and movement against a rough surface. Affected areas are usually discolored. Inspect sling and remove from service if core yarn is exposed.

TO PREVENT
Never drag slings along the ground. Never pull slings from under loads that are resting on the sling. Use wear protection between slings and rough surface loads.

SUGGESTED CORE-TEX PRODUCT: Wear Pad
Contact Core Lifting for more sling protection options.